

Research papers

Efficient energy management for renewable energy integrated multi-source microgrid with electric vehicle using Portia spider algorithm and temporal dynamic graph neural network technique

A. Madhan^{a,*}, A. Shunmugalatha^b, A.S. Vigneshwar^c

^a Assistant Professor, Department of Electrical and Electronics Engineering, Velammal College of Engineering and Technology, Madurai, Tamilnadu, India

^b Dean (Academics) & Professor, Department of EEE, Velammal College of Engineering and Technology, Madurai, Tamilnadu, India

^c Assistant Professor, Department of Electrical and Electronics Engineering, Ramco Institute of Technology, Rajapalayam, Tamil Nadu, India

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ABSTRACT

Microgrids based on renewable energy have gained prominence as environmental sustainability has received more attention globally. However, the fluctuating nature of renewable energy sources (RES), erratic energy consumption, and the dynamic influence of EVs on microgrid security pose a threat to their stability. This manuscript proposes a hybrid energy management strategy (EMS) for a multi-source microgrid that incorporates electric vehicles (EVs), RES, and an adaptive fractional order proportional-integral-derivative (AFOPID) controller to address these issues. The proposed approach combines the Portia Spider Algorithm (PSA) and Temporal Dynamic Graph Neural Network (TDGNN), which is termed the PSA-TDGNN method. The primary goal of the proposed method is to develop system efficiency, minimize grid dependency, minimize estimation error, and maintain a stable and reliable power supply for connected loads. The TDGNN is used to predict performance under high EV penetration levels, while the PSA is employed to optimize the AFOPID controller's gain factors. The proposed technique is assessed and compared utilizing the MATLAB platform to other existing approaches. The proposed approach yields better outcomes compared to existing methods like the Advanced-Sine Cosine Algorithm (a-SCA), Adaptive Salp Swarm Algorithm (ASSA), and Bald Eagle Search (BES). The proposed approach achieves an error of 0.59%, efficiency of 97%, and computational complexity of 18.23 s. These results demonstrate that the proposed technique achieves better performance and enhances energy management in multi-source microgrids based on EVs compared to existing methods.

1. Introduction

Global warming is a result of the exploitation of fossil fuel resources and the destruction of environmental conditions brought on by the sharp rise in the demand for electricity worldwide [1]. The transportation sector's growing dependence on fossil fuels makes it a major supplier to global greenhouse gas (GHG) emissions, along with the energy sector [2]. Due to the numerous socio-economic and environmental advantages that electric vehicles (EVs) offer, the governments of a number of nations are working to transform traditional transportation systems into green ones [3]. The electricity provided by the fossil fuel-based electrical distribution grid, however, indirectly raises the fossil fuel consumption of these EVs even though they do not directly use those [4]. The rapid adoption of EVs as a less expensive mode of transportation is further

hampered by a lack of charging facilities, especially in developing countries [5]. As a result, EV owners use the home connection to recharge their batteries, resulting in a low profitability index and a considerable system loss in the power sector [6]. A lot of EV chargers also have power quality concerns that are related to the distribution grid, like power loss, harmonics, and voltage swings due of their non-linear behavior [7]. These distribution network power quality issues are ascribed to disorganized and ineffective EV charging programs [8]. These can be resolved through energy management techniques, integrating renewable resources, enhancing converter topologies, and redesigning charging patterns [9]. From a techno-economic and environmental standpoint, using the renewable resources that are currently available is thought to be the best course of action. Additionally, it improves power quality by lessening the strain on the utility grid

* Corresponding author.

E-mail addresses: amadhan90@gmail.com (A. Madhan), asl@vcet.ac.in (A. Shunmugalatha), viigneshs@gmail.com (A.S. Vigneshwar).

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[10–13].

Renewable power-generation infrastructure has become increasingly important because of its low maintenance requirements, low cost, and environmental friendliness, despite its high capital requirement and intermittent nature. However, because renewable power generation is unpredictable and variable, integrating renewables for EV charging lowers system security and reliability. By using hybrid renewable resources for EV charging stations, these restrictions can be removed.

1.1. Literature survey

Many studies, according to EM, for a multi-source microgrid with electric vehicles based on renewable energy sources using different approaches and aspects can be found in the literature. Several of these reviews were followed.

Sahu et al. [14] have presented a fractional order fuzzy power system stabilizer (FO-Fuzzy PSS) that was employed in the control scheme for a number of dynamic situations. In the previously mentioned disturbances, the most promising method for achieving system frequency stability was the suggested FO-Fuzzy PSS control scheme. Under different conditions, the a-SCA was suggested as the best method for altering the controller gains. Regarding the microgrid system's frequency regulation, the optimum FO-Fuzzy PSS controller's performance was contrasted with that of the PID and standard fuzzy controllers.

Panda et al. [15] have suggested an ideal smart home energy management system (HEMS) that uses intelligent and dynamic load-priority scheduling. When it comes to a home EMS, the goals of priority-based scheduling were multi-focused and included cost-effectiveness in terms of electricity price savings, consumer choice of consumption based on priority, and peak load reduction. The DSM needs to include the concerns about RES uncertainties brought on by environmental dependency. The available sources, prioritized-based loads, and critical system factors have been modeled and optimized using the ASSA with a single objective discrete formulation for scheduling, ideally, the household appliance's operation.

Ed Ghadbane et al. [16] have introduced a lithium-ion battery-supercapacitor hybrid storage system that was ideal for the EMS of an electric vehicle. A synchronous reluctance motor was powered by an inverter, and bidirectional DC-DC converters connected the storage systems in parallel to the DC bus. By combining the best aspects of both methods, the suggested energy management strategy was based on fractional order integral sliding mode control and the BES algorithm. An urban dynamometer driving schedule (UDDS) driving cycle was considered to assess the efficacy of the recommended ideal energy management approach.

Zhang et al. [17] have developed the “1 + n” mixed platoon concept, which serves as the foundation for an effective nested parallel optimization (NPO) approach. With the approach, CHEV speed planning and energy management issues were simultaneously resolved using Pontryagin's smallest principle in a constrained optimum control framework. While taking into account fuel consumption and traffic efficiency across several intersections, it successfully shrinks the state and action space dimensions.

Kranthikumar et al. [18] have recommended a hybrid strategy for electric vehicle (EV) energy management that combines battery and super-capacitor technologies. The EMCABN-ROA technique was a hybrid strategy that associates the Remora Optimization Algorithm (ROA) with the developed Multi-Head Cross Attention driven Bidirectional Long Short Term Memory (Bi-LSTM) Network (EMCABN). The technique uses an EMCABN method for low-frequency battery power need prediction, which was a trained online using load power requirement to match the characteristics of hybrid energy storage datasets.

AlKawak et al. [19] have introduced the NBO-QNN hybrid methodology, which combines the Quantum Neural Networks (QNN) and Namib Beetle Optimization (NBO) techniques. The recommended energy management strategy decreases EV power usage and increases

battery life. To satisfy load needs, QNN anticipates and mixes power supply and charge levels. NBO was used to continuously control output voltage, current, and generate references in EV energy management. Higher energy density batteries and power density SC were able to meet vehicle requirements.

Mohammad et al. [20] have suggested aims to alleviate EV users' range anxiety and introduce a novel trading mechanism for transactive trading between workplace EVs. In comparison to a traditional V2G system, a commercial area in Auckland exhibits lower charging costs. Additional research reveals that using public charging stations can save money as well. However, their influence was small in comparison to the quantity of EVs that were discharging and engaging in transactive trading. The cost savings of the recommended approach were also demonstrated by PV generation uncertainty analysis in different scenarios.

Xiaoluan et al. [21] have developed a microgrid (MG) comprising various renewable energy-driven technologies, like wind turbines (WT), solar PV systems, and plug-in hybrid electric vehicles (PHEVs), was studied for its day-ahead functioning difficulty. The output power, price, loads, and electric car charging demand uncertainties of the stochastic RERs were all taken into consideration while modeling the suggested day-ahead scheduling for a whole day. The purpose of the problem's execution was to maximize the MG's total operating expenses. The high complexity of the problem was solved by applying the hybrid crow search and pattern search (HCS-PS) method, which considers the entire search space globally.

1.2. Research gap and motivation

The generic review of current work highlights that energy management in multi-source microgrids integrating electric vehicles is an essential area of study. The simultaneous optimization of microgrid stability, cost-effectiveness, and energy storage efficiency is a challenging task. The difficulty arises because of intermittent renewable energy generation, varying load demand, and the stochastic nature of electric vehicle charging and discharging cycles. Many researchers have addressed energy management challenges using various optimization techniques such as a-SCA, ASSA and BES. The a-SCA technique struggles with slow convergence and local optima issues in complex optimization problems. The ASSA approach faces challenges related to uncertainty in RES and dynamic demand-side management. The BES algorithm improves efficiency but lacks robustness in handling multiple conflicting objectives simultaneously. Moreover, existing research predominantly applies single-objective control methods, lacking a unified framework that integrates multiple advanced techniques. To address these limitations, this research proposes an intelligent technique to enhance efficiency, improve system reliability, and optimize power flow in microgrids with electric vehicles. The proposed approach aims to bridge existing gaps by providing a scalable, robust, and intelligent energy management solution for future smart grids.

The novelty of this research lies in the integrated use of the AFOPID controller with the Portia Spider Algorithm (PSA) and Temporal Dynamic Graph Neural Network (TDGNN) within a unified framework for energy management in multi-source microgrids with high EV penetration. Unlike conventional approaches that treat optimization and prediction separately, this method simultaneously tunes the AFOPID controller gains using PSA while forecasting spatiotemporal variations in renewable generation, load demand, and EV charging through TDGNN. This integration generates a synergistic effect: predictive insights from TDGNN guide the PSA-optimized AFOPID controller to adaptively manage energy flows, improving battery utilization, minimizing control error, and maintaining stable load balance under highly dynamic conditions. By combining adaptive control, swarm intelligence optimization, and dynamic graph-based prediction, the PSA-TDGNN approach enhances system stability, operational robustness, and computational efficiency, outperforming traditional swarm intelligence

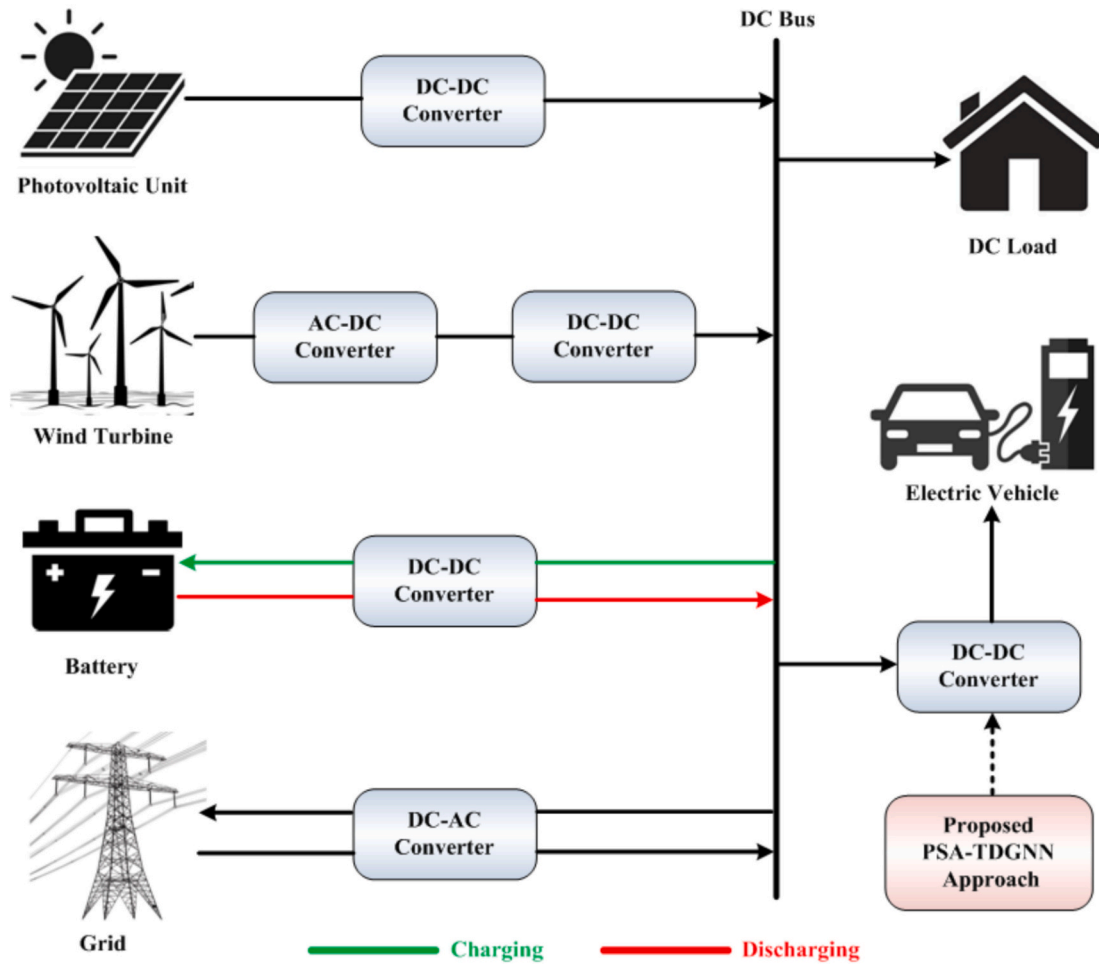


Fig. 1. Proposed EV energy management system's structure.

or deep learning (DL) approaches in terms of accuracy, energy efficiency, and real-time applicability.

1.3. Contribution

The major contributions of the paper are as follows:

- The novel PSA-TDGNN method is proposed for energy management for a multi-source microgrid integrated with RES, EVs and AFOPID controller.
- The AFOPID controller generalizes the conventional PID by incorporating fractional calculus in the integral and derivative terms, and is employed to enhance stability and adaptability under nonlinear renewable and EV load dynamics.
- The PSA is employed to optimally tune the gain factors of an AFOPID controller, improving system dynamic response and reducing estimation error under fluctuating renewable and EV load conditions.
- The TDGNN is leveraged to model temporal dependencies and predict system performance under high EV penetration, ensuring stability and proactive management of microgrid operations.
- The significance of the proposed PSA-TDGNN approach lies in its capacity to improve energy efficiency and reliable power supply in multi-source microgrids by optimizing energy distribution and predicting system performance under high EV penetration.

The balance of the paper is ordered as follows: The setup of a grid-connected PV/wind conversion system with an EV is described in part 2. Part 3 displays the proposed PSA-TDGNN. The results and discussion

are explained in part 4. Part 5 concludes the paper.

2. Configuration of grid-connected PV/wind conversion system with electric vehicle

Fig. 1 demonstrates the structure of the proposed EV EMS in a renewable-integrated multi-source microgrid, where the DC bus serves as the central hub for energy flow and power distribution. While the WT produces AC power that is first rectified with an AC-DC converter and then stabilized with a DC-DC converter to ensure seamless integration, the PV unit generates DC power that is processed through a DC-DC converter before being supplied to the bus. A bi-directional DC-DC converter connects the battery storage system (BSS), allowing it to store extra energy throughout times of excess generation and return power to the bus when demand outpaces renewable output. A DC-AC converter creates the grid connection, enabling the system to either deliver excess power when generation surpasses local consumption or draw energy during shortages. The EV system is linked to the DC bus with a DC-DC converter, enabling both charging and vehicle-to-grid discharging functions, thereby positioning EVs as flexible energy assets within the microgrid. At the control layer, the proposed PSA-TDGNN approach intelligently manages interactions among RES, battery, EVs, and the grid, the PSA optimizes the gain parameters of the AFOPID controller, while the TDGNN predicts dynamic performance under varying load and EV penetration conditions. This integrated control strategy ensures a reliable power supply, improved efficiency, reduced grid dependency, and enhanced system stability, making the framework adaptive and sustainable for future smart energy systems.

2.1. Modeling of electric vehicle

The coordination of RESs in the microgrid is greatly impacted by EVs. EVs can be used for power distribution network voltage management by carefully controlling their charge rate, SoC, and synchronized charging–discharging [22]. The present study assessed the effect of EVs on objective functions with a controlled charging technique. According to the vehicle's battery's SoC, the EV charging strategy facilitates the charging of EVs. The distance traveled and the energy consumed by an EV determines the available energy in its battery, which is represented as follows:

$$SOC = 1 - \frac{d}{R} \quad (1)$$

$$SOC_{charge,n} = 1 - SOC_n \quad (2)$$

$$SOC_{discharge,n} = SOC_n - SOC_{EV}^{min} \quad (3)$$

where, SOC_n is indicated as the SoC of the n^{th} EV battery, d is indicated as the distance traveled, R is the maximum driving range, $SOC_{charge,n}$ is denoted as the required SoC for charging of the n^{th} EV, $SOC_{discharge,n}$ is denoted as the discharging SoC of the n^{th} EV, SOC_{EV}^{min} is indicated as the actual SoC level of the n^{th} EV. The following represents the whole discharging and charging durations of the n^{th} EV at a charging station:

$$t_{charge,n} = \frac{SOC_{charge,n} \times S \times 60}{P_{charge,n}} \quad (4)$$

$$t_{discharge,n} = \min\left(\frac{SOC_{discharge,n} \times S \times 60}{P_{discharge,n}}, 1440 - t_{start,n}\right) \quad (5)$$

where, $t_{charge,n}$ is denoted as the charging time of the n^{th} EV, S is denoted as the battery capacity, $t_{discharge,n}$ is indicated as the discharging time of the n^{th} EV, $P_{charge,n}$ is indicated as the charging power of the n^{th} EV, $P_{discharge,n}$ is denoted as the discharging power of the n^{th} EV. To accurately represent the EV system, the charging configuration is modeled with constant current (CC) mode reference current (i_{b1ref}) of 40 A and constant voltage (CV) mode reference voltage (V_{b1ref}) of 500 V. The EV battery parameters are defined with a nominal capacity (Q_n) of 50 Ah, an internal resistance (R_{s1}) of 100 mΩ, a polarization resistor (R_{p1}) of 1 μΩ, and a polarization capacitor (C_{p1}) of 500 F. These parameters ensure that the charging-discharging dynamics of the EV are realistically captured for integration within the microgrid.

2.2. Modeling of photovoltaic system

Electricity is produced by the PV panel using solar energy. The majority of the solar radiation that reaches PV panels is transformed into heat, with only a small portion being transformed into electricity [23]. Therefore, both solar irradiance and temperature effects are considered when modeling PV output. An individual PV panel's output power is expressed as:

$$P_{sol}(t) = \frac{P_m S_m}{1000} [1 - \lambda(T_{cell} - 25)] \quad (6)$$

where, $P_{sol}(t)$ is indicated as the output power of the n^{th} PV panel at period t , T_{cell} is indicated as the cell temperature of PV panel, λ is indicated as the temperature coefficient of power, S_m is indicated as the solar radiation, P_m is the rated maximum power. The cell temperature can be computed using the following formula, which depends on the surrounding temperature and irradiance:

$$T_{cell} = T_{amb} + \frac{NOCT - 20}{0.8} S_m \quad (7)$$

here, T_{amb} is indicated as the ambient temperature, $NOCT$ is denoted as

the temperature of the cell, which is the nominal operating temperature when T_{amb} is 20 °C and solar irradiance is 0.8 kW/m². The rated maximum power P_m of each PV panel is calculated as:

$$P_m = V_m I_m \quad (8)$$

where, V_m is the maximum voltage of the n^{th} PV panel, I_m is indicated as the maximum current of the n^{th} PV panel. Finally, the total power generated by all PV panels in the system at moment t is:

$$P_{PV}(t) = N_{PV} P_{sol}(t) \quad (9)$$

where, $P_{PV}(t)$ is indicated as the total power created by PV panels at time t , N_{PV} is denoted as the number of PV panels in the system. PV system parameters include a cell temperature of 60 °C, a maximum power of 243.4 W, a temperature coefficient of 0.07, a short-circuit current of 8.81 A, an open-circuit voltage of 38.3 V, a voltage at maximum power of 30.2 V, and a current at maximum power of 8.06 A.

2.3. Modeling of wind turbine

The mechanical power extracted from a wind turbine depends on wind speed, blade radius, and aerodynamics [24]. The mechanical torque (T_m) created by the wind turbine is given by:

$$T_m = 0.5 \rho \pi R^3 C_p V^2 / T_{SR} \quad (10)$$

where, ρ is indicated as the air density, R is indicated as the radius of the turbine blade, C_p is denoted as the power coefficient, representing the efficiency of wind energy conversion, V is represented as the wind speed, π is denoted as a constant, T_{SR} is the ratio of tip-speed. The ratio of tip-speed (T_{SR}) is defined as:

$$T_{SR} = \frac{\omega_m R}{V} \quad (11)$$

where, R is indicated as the blade radius (m), ω_m is denoted as the turbine rotor's angular velocity (rad/s), V is denoted as the wind speed (m/s). The following empirical relation can be used to approximate the coefficient of power (C_p), which is dependent on the ratio of tip-speed and the blade pitch angle:

$$C_p = 0.5176 \left(\frac{116}{(\rho V / R + 0.08\beta) - (0.035/\beta^3 + 1)} - 0.4\beta - 5 \right) e^{-21\tau} \quad (12)$$

$$\tau = (\rho V / R + 0.08\beta) - (0.035/\beta^3 + 1)(0.0068\rho V / R) \quad (13)$$

where, β is denoted as the blade pitch angle, τ is indicated as the intermediate variable used in the calculation. The wind turbine is modeled with a nominal power of 30 kW, turbine radius of 1.3 m, blade pitch of 2°, nominal speed of 60 rad/s, rotor flux of 0.8 Wb, total inertia of 0.01 Nm·rad/s², and total viscous friction of 0.01 Kg·m²·s⁻¹.

2.4. Modeling of battery

The Battery Energy Storage System (BESS) are installed in the microgrid that is connected to PV systems. Peak shaving, minimizing variations in grid demand, and managing uncertain EV loading are all supported by the batteries. Deep-cycle lead-acid batteries with valve regulation are employed in this investigation [25]. The dynamic energy state of the battery can be modeled as:

$$E_b(t+1) = E_b(t) + \Delta t(P_b(t) - \eta_b |P_b(t)|) \quad (14)$$

where, $E_b(t+1)$ is denoted as the battery's stored energy at each time step, positive for charging, negative for discharging, η_b is denoted as the battery efficiency, $P_b(t)$ is represented as the electric power of the battery at time t (kW), Δt is denoted as the time step duration, $E_b(t)$ is denoted as the energy stored in the battery at time t . The initial energy

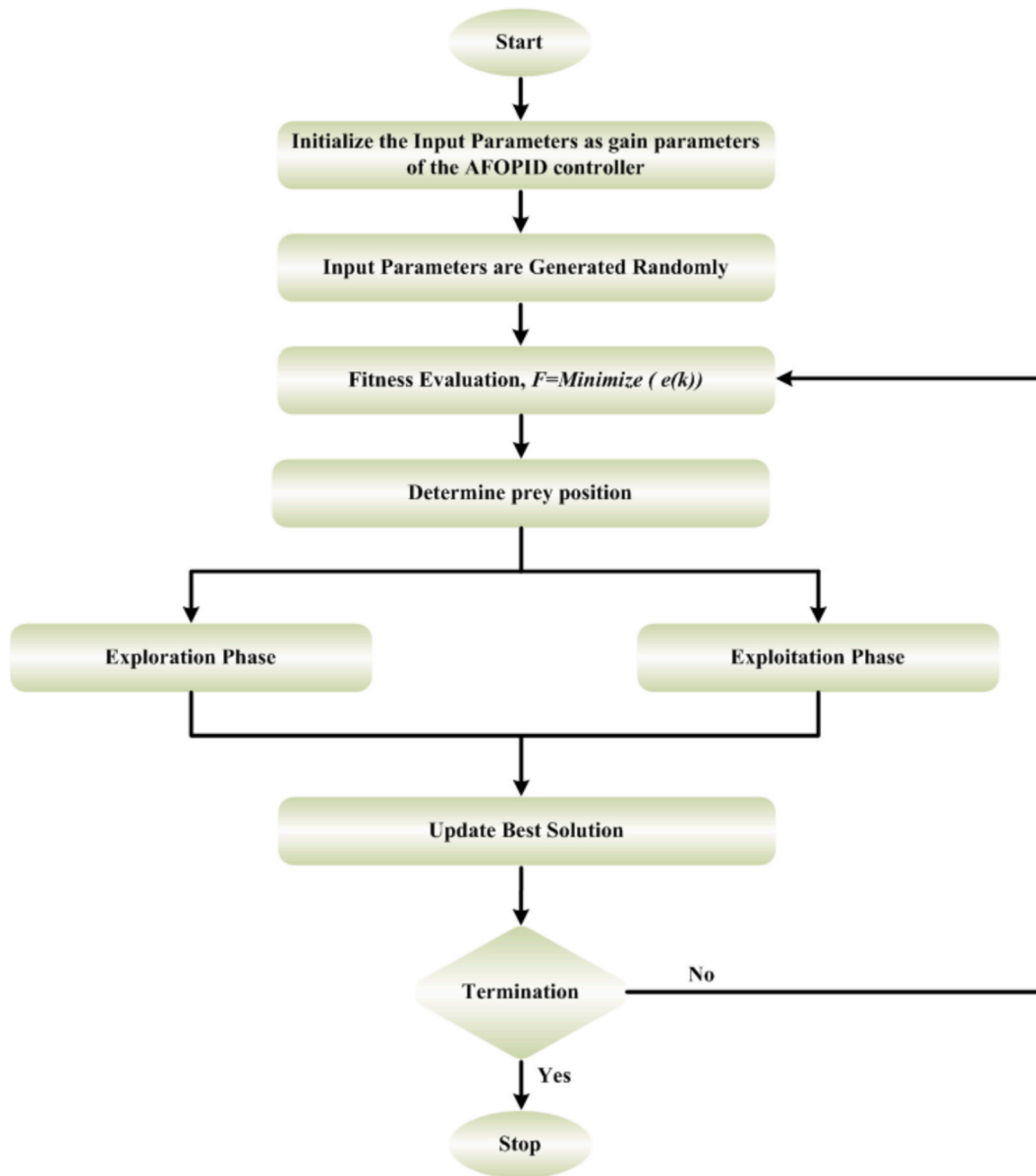


Fig. 2. Flowchart of PSA.

stored in the battery is:

$$E_b(0) = E_b(\text{initial}) \quad (15)$$

where, $E_b(\text{initial})$ is the battery's initial energy level (kWh). The SOC of the battery (SOC_b) is presented in Eq. (16).

$$SOC_b(t) = E_b(t) / (N_b \cdot Q_{b, \text{cap}}), \quad t = 0, \dots, N-1 \quad (16)$$

where, $Q_{b, \text{cap}}$ is the capacity of an individual battery unit (kWh), N_b is denoted as the count of batteries deployed in the system (optimized), $E_b(t)$ is indicated as the total energy stored in the battery bank at time t (kWh). To accurately represent the battery system, the parameters are defined with a CC mode reference current (i_{b2ref}) of 30 A, a CV mode reference voltage (V_{b2ref}) of 612 V, a nominal capacity (Q_n) of 50 Ah, an internal resistance (R_{s2}) of 100 mΩ, a polarization resistor (R_{p2}) of 1 μΩ, and a polarization capacitor (C_{p2}) of 500 F. These specifications ensure realistic modeling of the battery performance under charging and discharging conditions within the microgrid.

2.5. Adaptive fractional order proportional-integral-derivative controller (A-FOPID)

Fractional calculus and adaptive control are combined in an advanced control system called an A-FOPID controller to improve system performance [26]. FOPID is an expansion that Podlubny proposed to increase the efficiency of PID controller design. The fractional differential integral is constructed utilizing the Riemann-Liouville function definition, found in Eq. (17).

$${}_a D_t^\alpha F(t) = \frac{1}{\Gamma(n-\alpha)} \left(\frac{d}{dt} \right)^n \int_a^t \frac{f(\xi)}{(t-\xi)^{1-(n-\alpha)}} d\xi \quad (17)$$

here, D_t^α is the fractional derivative/integral operator of order α , $F(t)$ is the function to be differentiated/integrated, $\Gamma(\cdot)$ is Euler's gamma function, generalizing the factorial to non-integer values. Alternatively, the Grünwald-Letnikov definition can approximate fractional differentiation as:

Table 1

Pseudocode of PSA.

Input parameters: Gain parameters of AFOPID controller such as K_p , K_i , K_d , λ and μ .
 Begin
 Create random Portia spiders;
 while ($I_{cur} < I_{max}$) do
 Compute the fitness value;
 Determine prey location;
 Exploration phase
 Update value of alpha 1;
 Compute standardized fitness score;
 Update Portia spider location;
 Exploitation phase
 Update value of alpha 2, alpha 3, alpha 4, and alpha 5;
 Update Portia spider utilizing location;
 end
 Update Best Solution;
 Determine the best solution;
 end
 Output: Optimal solution.

$$\alpha D_t^\alpha F(t) = \lim_{g \rightarrow 0} \frac{\Gamma(\alpha + d)}{(g + 1)} f(t - dg) \quad (18)$$

where, d is the discrete time step. Remember that the fractional-order operator design can be used to consolidate the integrator. Using the Laplace transform, the transfer function and time-domain control law of the FOPID controller is expressed as:

$$G_c(s) = K_p + K_i s^{-\lambda} + K_d s^\mu \quad (19)$$

$$u(t) = K_p e(t) + K_i D_t^{-\lambda} e(t) + K_d D_t^\mu e(t) \quad (20)$$

here, $G_c(s)$ is I as the FOPID controller transfer function, K_p is indicated as the proportional gain, K_d is indicated as the derivative gain, K_i is indicated as the integral gain, μ is denoted as the order of fractional differentiation, λ is indicated as the order of fractional integration, $u(t)$ is denoted as the controller output signal, $D_t^{-\lambda} e(t)$ is denoted as the fractional-order integral of error, $D_t^\mu e(t)$ is denoted as the fractional-order derivative of error. The AFOPID controller parameter bounds were set as $K_i \in [0, 100]$, $K_p \in [0, 10]$, $K_d \in [0, 10]$, $\lambda \in [0.1, 1.5]$, and $\mu \in [0.1, 1.5]$.

2.6. Objective function

To develop stability and performance, the objective function is made to reduce error in microgrid parameter control [27]. The error in discrete time, along with its variation, is used as the input variable for the controller, and its output is shown as:

$$e(k) = x^*(k) - x(k) \quad (21)$$

here, $e(k)$ is the error, $x(k)$ is denoted as the process output variable, and $x^*(k)$ is indicated as the desired reference signal.

3. Energy management of multi-source microgrid with electric vehicle using proposed PSA-TDGNN approach

In this study, a novel PSA-TDGNN approach is proposed for energy management in multi-source microgrids integrating RES and electric vehicles. The proposed method combines an AFOPID controller for robust and adaptive energy control, the PSA for optimal tuning of controller gains, and a TDGNN for predicting spatiotemporal variations in renewable generation, load demand, and EV charging/discharging patterns. Existing methods, such as the a-SCA, ASSA, and BES suffer from slow convergence, limited ability to handle multiple conflicting objectives, and insufficient predictive performance under fluctuating RES output and dynamic EV behavior. Unlike these approaches, the PSA-TDGNN framework integrates optimization and predictive modeling,

enabling proactive energy flow management, improved battery utilization, reduced grid dependency, and reliable load balancing. By coupling predictive insights with adaptive control, the proposed method achieves higher system efficiency, lower computational complexity, and more stable microgrid operation, providing a scalable and robust solution for intelligent energy management in modern multi-source microgrids with high EV penetration. A brief description of the proposed PSA-TDGNN techniques is given in following part:

3.1. Optimizing the AFOPID controller's gain parameters using the Portia spider algorithm

In this section, the optimization of the AFOPID controller's gain factors utilizing the Portia Spider Algorithm (PSA) is discussed [28]. The Portia spider's strategic and predatory behavior served as the model for the PSA, a new swarm intelligence algorithm. To reduce the control error in the microgrid energy management system, PSA is used in this study to optimize the five important AFOPID controller parameters: derivative gain K_d , integral order λ , proportional gain K_p , integral gain K_i , and derivative order μ . The flowchart of the PSA is denoted in Fig. 2. The pseudocode of PSA is shown in Table 1. The steps of PSA are given below;

Step 1: Initialization

Set the input factor; here, the inputs are gain factors of the AFOPID controller, like K_p , K_i , K_d , λ and μ .

Step 2: Random Generation

Randomly create initial candidate solutions within the allowable range of each gain parameter;

$$S = \begin{bmatrix} S_{1,1} & S_{1,j} & S_{1,m} \\ S_{i,1} & S_{pi,j} & S_{i,m} \\ S_{N,1} & S_{N,j} & S_{N,m} \end{bmatrix} \quad (22)$$

here, S is represented as the population matrix of spiders locations, N is represented as the count of Portia spider, mis represented as the parameters related to the problem of optimization, $S_{i,j}$ is the j^{th} parameter of the i^{th} spider.

Step 3: Fitness Function

The fitness of each candidate solution is evaluated using the control error.

$$F = \text{Minimize}(e(k)) \quad (23)$$

here, F is denoted as the fitness value of a candidate solution, $e(k)$ is indicated as the control error at time step k . The candidate with the lowest error is considered the best solution.

Step 4: Determine Prey Position

In AFOPID gain optimization, the prey represents the controller parameters yielding the lowest control error. At each iteration, candidate solutions compare their errors to the prey, and the one with the minimum error becomes the new prey, guiding the swarm toward better solutions. Mathematically, the prey position vector is defined as;

$$T = [t_1 t_2 \dots t_d] \quad (24)$$

where, t is the prey vector representing the current best solution and dis the dimension of the solution vector.

Step 5: Exploration Phase

The exploration phase in AFOPID gain optimization allows candidate solutions to search widely across the parameter space, perturbing their parameters randomly to avoid local minima and find gain combinations that minimize tracking error and optimize battery SoC. Mathematically, the updated parameter for j^{th} gain of the i^{th} candidate is given by:

$$S_{ij}^{t+1} = \begin{cases} t_j & \text{if } \alpha_1 < SF(S_i) \\ s_{ij}^t & \text{if } \alpha_1 \geq SF(S_i) \end{cases} \quad (25)$$

$$SF(S_i) = \frac{F(S_i)}{\sqrt[2]{\sum_1^N F(S_i)^2}} \quad (26)$$

where, S_{ij}^{t+1} is the updated value of the j^{th} gain parameter for the i^{th} spider, α_1 is the random factor controlling exploration, $SF(S_i)$ represents the fitness of the candidate based on microgrid performance metrics, t_j represents the j^{th} factor of the prey, s_{ij} denotes the j^{th} parameter of the i^{th} solution.

Step 6: Exploitation Phase

The exploitation phase in AFOPID optimization refines gain parameters in promising regions, with candidate solutions moving toward the best-known solution to minimize control error, improve load balancing, and maintain voltage stability under varying renewable generation and EV charging. Mathematically, the parameter update is expressed as;

$$S_{ij}^{t+1} = t_j^t + \left| \alpha_2 t_j^t - s_{ij}^t \right| \times e^{\alpha_3} \times \cos(2\pi\alpha_4) \quad (27)$$

$$S_{ij}^{t+1} = t_j^t + \alpha_2 \times \left(t_j^t - s_{ij}^t \right) \quad (28)$$

where α_2 and α_4 are denoted as arbitrary values in the interval $[0, 1]$, t_j is the prey's j^{th} parameter, and α_3 is established by making sure the exploration and exploitation stages are balanced.

Step 7: Update Best Solution

The best-performing AFOPID gains are updated by combining damping effects with solution adjustments, guiding convergence toward optimal values while maintaining swarm diversity to avoid local minima. This ensures minimal control error, stable voltage, and efficient energy management under varying renewable generation and EV charging. The updated best solution is mathematically expressed as:

$$S_{ij}^{\text{best}} = \begin{cases} t_j^t + |\alpha_2 t_j^t - s_{ij}^t| \times e^{\alpha_3} \times \cos(2\pi\alpha_4) \alpha_5 < 0.5 \\ t_j^t + \alpha_2 \times (t_j^t - s_{ij}^t) \alpha_5 \geq 0.5 \end{cases} \quad (29)$$

here, S_{ij}^{best} is denoted as the updated best position of the i^{th} spider for parameter j , α_5 is the additional random factor to maintain diversity.

Step 8: Termination

Check the conclusion criteria; if the optimal AFOPID controller gains $[K_p, K_i, K_d, \lambda, \mu]$ achieving the minimum control error are found, or the maximum count of iterations is reached, the process stops. Otherwise, repeat the optimization steps until convergence.

After optimizing the AFOPID controller gains through PSA to enhance energy flow control, the system's ability to handle varying loads and high EV penetration is further evaluated using the TDGNN, which predicts microgrid performance under dynamic operating conditions.

3.2. Predicting the performance under high EV penetration using temporal dynamic graph neural network

After optimizing the AFOPID controller gains through PSA, the microgrid's performance under high EV penetration is predicted utilizing a TDGNN that captures both spatial and temporal dependencies among key components [29], including PV arrays, wind turbines, batteries, loads, EV chargers, and the grid interface. The microgrid is modeled as a dynamic graph, with nodes representing physical components:

$$V = \{PV_1, PV_2, WT_1, Batt_1, EV_1, Load_1, \dots\} \quad (30)$$

Edges represent interactions based on either physical impedances or power-flow coupling, forming a weighted adjacency matrix A . The initial adjacency matrix is constructed as:

$$A = \Theta^T \cdot \Psi \quad (31)$$

where, and Θ are learnable node embedding vectors for each time slot. The edge weights are normalized, and the adjacency matrix is symmetrically normalized. Node embeddings are updated using a Dynamic Graph Isomorphism Network that propagates information across neighbors and incorporates temporal information from the previous time step:

$$h_v^{(l,t)} = MLP^{(l,t)} \left(\left(1 + A^{(l)} \right) \cdot h_v^{(l-1,t)} + h_v^{(l-1,t-1)} + \sum_{u \in N(v)} \bar{w}_{ij} \cdot h_u^{(l-1,1)} \right) \quad (32)$$

where, $\bar{w}_{ij}$ are the normalized edge weights, $A^{(l)}$ is a learnable parameter, and $h_v^{(l-1,t)}$ incorporates temporal information from the previous time step. Embeddings across all time steps are concatenated to form the temporal representation:

$$h_v^{(l)} = CONCAT(h_v^{(l,t)} | t = 0, 1, \dots, T) \quad (33)$$

A temporal convolution layer is then applied to extract local temporal patterns from each node's embedding, ensuring that both spatial and temporal dependencies are captured before prediction:

$$X^{l+1} = \sum_{j=0}^{N^l-1} weight(N^{l+1}, j) * X^l + bias(N^{l+1}) \quad (34)$$

where, $*$ is denoted as the valid 2D cross-correlation operator, N indicated as the number of nodes, $weight(\cdot)$ and $bias(\cdot)$ are the convolution kernel parameters. The TDGNN is trained using a regression loss appropriate for forecasting, specifically the Mean Squared Error (MSE) with regularization:

$$Loss = \sum_S \sum_{i=1}^{|I|} \hat{y}_N \log \left(\hat{y}_{N(i)} \right) + (1 - y_N) \log \left(1 - \hat{y}_N \right) + \lambda \|\Theta\|_2 \quad (35)$$

where, y_N is the ground truth power, Θ denotes all learnable parameters, λ is the regularization strength. The Adam optimizer is used to update all learnable parameters including node embeddings, edge weights, and convolution kernels.

This dynamic graph formulation allows the TDGNN to anticipate fluctuations in renewable generation, load demand, and EV charging/discharging patterns, enabling proactive strategies such as optimal battery dispatch, load balancing, and reduced grid dependency. The PSA-TDGNN framework is computationally efficient, scalable, and responsive, with prediction and optimization times well within microgrid control intervals, making it feasible for real-time energy management under high EV penetration. The parameter setting of the PSA-TDGNN approach is demonstrated in Table 2.

Table 2
Parameter setting of PSA-TDGNN Approach.

Parameters	Values
PSA	
Population size	50
Search agents	25
Iterations	100
Random value	[0, 1]
TDGNN	
Pooled ratio	0.2
Epochs	2000
Batch size	16
Learning rate	10^{-4}
Random Seed	42/2024
Early Stopping	Patience = 50 epochs

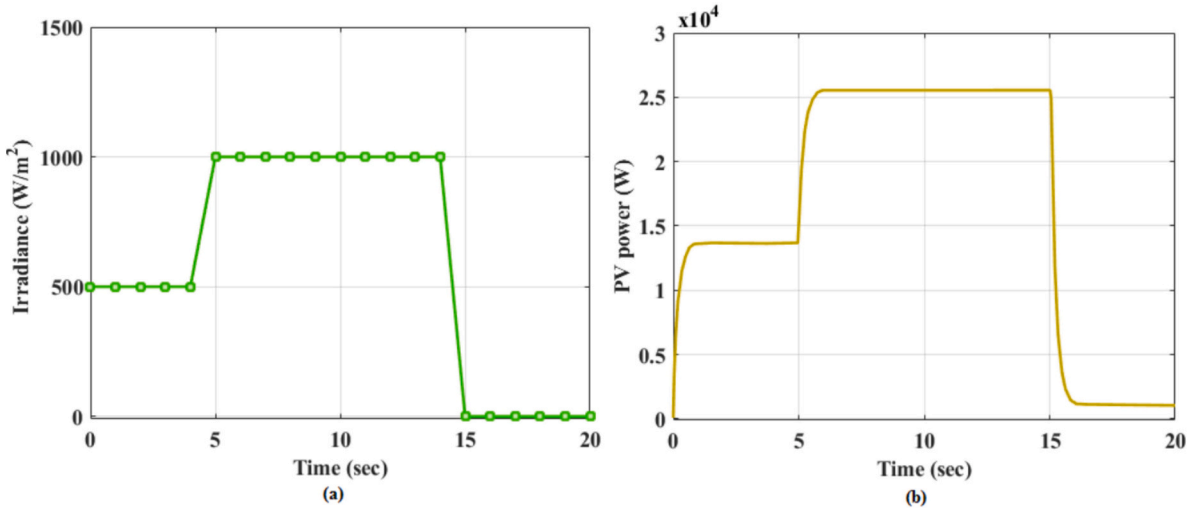


Fig. 3. Analysis of photovoltaic performance (a) Irradiance (b) PV power.

4. Result and discussion

This part gives the performance of the proposed PSA-TDGNN technique based on simulation results, with emphasis on EV charging scenarios in a multi-source microgrid. Simulations were conducted for both grid-connected (grid ON) and islanded (grid OFF) operating modes. The framework was implemented and validated on MATLAB 2023a and benchmarked against established optimization techniques. The a-SCA, ASSA, and BES algorithms were selected as comparison methods because they are widely used swarm intelligence optimizers in microgrid energy management, providing suitable reference points for evaluating the effectiveness of the proposed approach. A system with an Intel(R) Core(TM) i5-3230M CPU @ 2.60 GHz, 4 logical processors, 2601 MHz, 2 cores, and 8 GB of RAM running Microsoft Windows 10 Pro, version 10.0.19045 Build 19045, was used for the simulations. For training and evaluating the TDGNN, simulated datasets were divided into 75% for training and 25% for testing. The model's predictive ability under high EV penetration scenarios during charging was rigorously evaluated utilizing standard performance metrics, like prediction accuracy, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R2). These specifications, combined with the processor, memory configuration, and software environment, demonstrate the computational feasibility and real-time applicability of the proposed PSA-TDGNN framework. The simulation of the proposed

PSA-TDGNN method is given below;

The analysis of photovoltaic performance is illustrated in Fig. 3. Subplot 3(a) illustrates the irradiance profile, which rises from 500 W/m² to 1000 W/m² within the first 5 s, remains stable, and drops to zero at 15 s. Subplot 3(b) shows the corresponding PV output power, which closely follows the irradiance trend, confirming the model's realistic response under dynamic environmental conditions. The analysis of the wind energy system is illustrated in Fig. 4. Subplot 4(a) presents the wind speed, which increases from 6 m/s to 15 m/s, holds steady, and then declines to nearly zero at 15 s. Subplot 4(b) demonstrates the resulting wind turbine power output, which rises sharply with wind speed and diminishes as the wind subsides, validating the system's capability to capture fluctuating wind energy. The analysis of battery SOC during charging and discharging cycles is displayed in Fig. 5. The SOC profile increases steadily from below 99.9% to full charge at 10 s, remains constant until 15 s, and then decreases due to discharging. This behavior confirms that the BESS efficiently manages charging and discharging cycles. The analysis of DC link voltage performance is displayed in Fig. 6. The DC link voltage rapidly stabilizes at around 700 V and maintains this level with minimal fluctuation for the entire 20-second duration, indicating strong voltage regulation and system stability. The analysis of power factor correction with current waveform is displayed in Fig. 7. The current waveform oscillates between 200 A and -250 A within the observed time frame, reflecting effective power factor

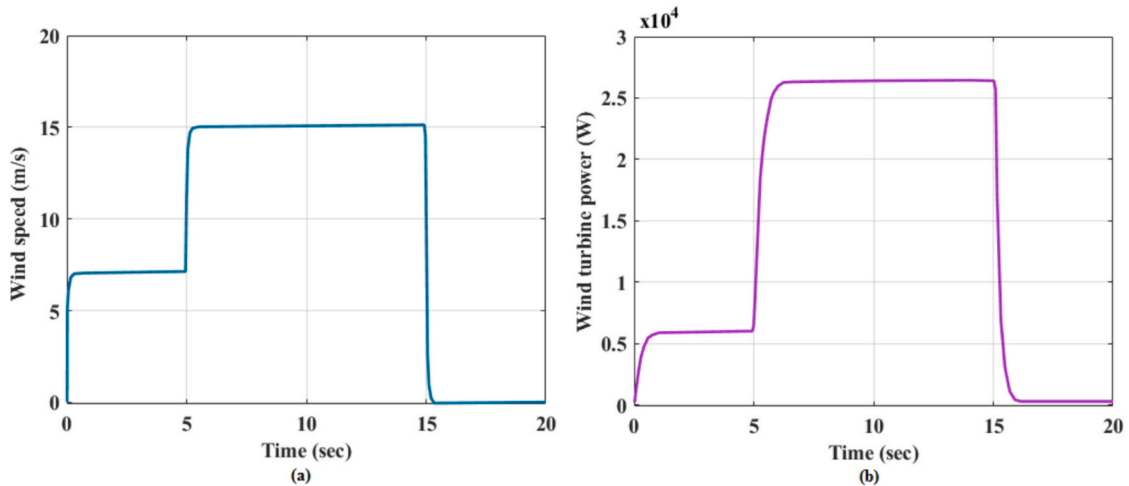


Fig. 4. Analysis of wind energy system (a) Speed (b) WT power.

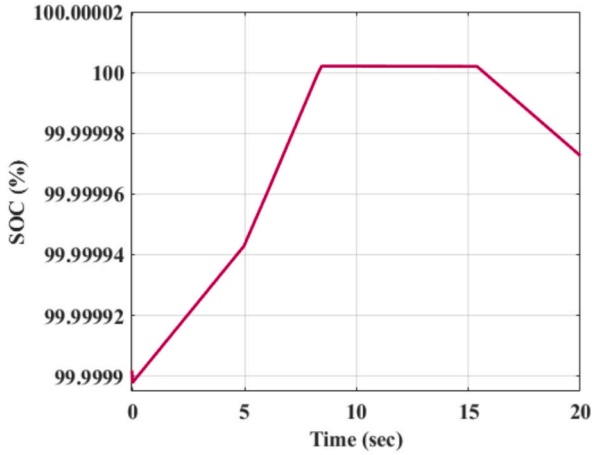


Fig. 5. Analysis of battery SOC during charging and discharging cycles.

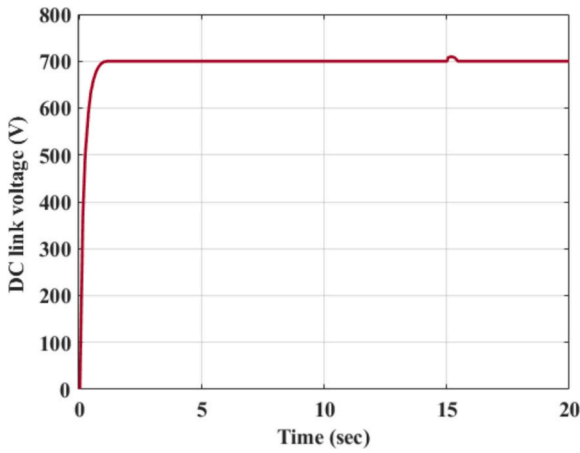


Fig. 6. Analysis of DC link voltage performance.

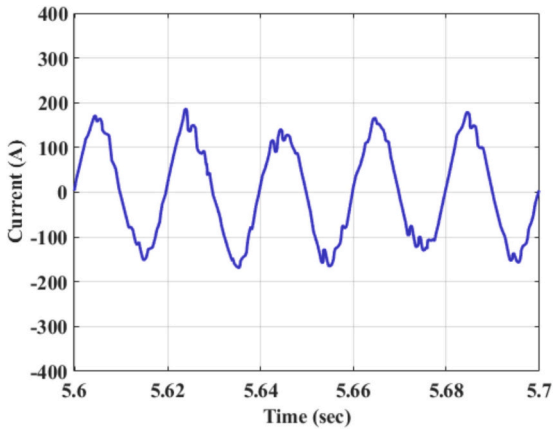


Fig. 7. Analysis of power factor correction with current waveform.

correction and reduced harmonic distortion. The analysis of battery operational characteristics is illustrated in Fig. 8. Subplot 8(a) displays that the battery current peaks near 35 A during charging and decreases to around -10 A during discharging. Subplot 8(b) indicates that the battery voltage rises to 615 V, stabilizes, and then slightly decreases, demonstrating a dynamic but controlled response to load variations. The performance analysis of the energy flow management system with a battery during EV charging is displayed in Fig. 9. Subplot 9(a) displays the EV charging while the grid is ON, where the battery steadily supplies

about 2×10^4 W before returning to zero, demonstrating its auxiliary support. Finally, subplot 9(b) illustrates EV charging with the grid OFF, where the battery power fluctuates around 2×10^4 W before declining, reflecting its critical role in sustaining charging during islanded operation.

The performance analysis of the energy flow management system with PV is displayed in Fig. 10. Subplot 10(a) presents EV charging with the grid ON, where PV power remains steady at about 2.5×10^4 W, easing grid demand. In subplot 10(b), during EV charging with the grid OFF, PV power reaches up to 22×10^4 W, fluctuates slightly, demonstrating its capacity to sustain charging independently. The performance analysis of the energy flow management system with load is displayed in Fig. 11. Subplot 11(a), under EV charging with grid ON, shows load demand fixed at 1×10^4 W, unaffected by charging. Finally, subplot 11(b), with EV charging and grid OFF, shows load demand around 1.8×10^4 W, reflecting higher reliance on local resources. The performance analysis of the energy flow management system with wind power is displayed in Fig. 12. Subplot 12(a), during EV charging with grid ON, shows wind power steady at 2.5×10^4 W, supporting charging and load. In subplot 12(b), with EV charging and grid OFF, wind power peaks at 2.3×10^4 W and gradually falls, demonstrating its contribution in isolated operation. The performance analysis of the energy flow management system with EV charging is displayed in Fig. 13. In subplot 13(a), with the grid ON, EV charging power increases to nearly 2×10^4 W before returning to zero, reflecting smooth, grid-supported charging. In subplot 13(b), with the grid OFF, EV charging power starts slightly negative, rises to 2×10^4 W with minor fluctuations, and then stabilizes, showing controlled standalone charging. The performance analysis of the energy flow management system grid power is displayed in Fig. 14. In subplot 14(a), with the grid ON, power shows slight fluctuations but stabilizes at around 4.5×10^4 W after 15 s, confirming reliability. In subplot 14(b), with the grid OFF, grid power steadily decreases to 0×10^4 W between 15 and 20 s, showing a stable transition to islanded operation.

Table 3 shows a comparison of the proposed method with existing approaches in terms of error, efficiency, and computational complexity. The proposed method achieves the lowest error of 0.59%, the highest efficiency of 97%, and the fastest computation time of 18.23 s. The P -values indicate that the improvements over a-SCA, ASSA, and BES are statistically significant, confirming the superior performance of the proposed approach across all evaluated metrics.

Table 4 indicates that the proposed PSA-TDGN approach is superior to previous hybrid methods and that it recorded the highest battery SoC of 98.52% and robustness of the system of 87% as compared to SoCs of 45.09%, 55.47%, and 81% reported by other methods and robustness statics of 68%, 70%, 72%, 80% from strategy performance and each 70–82% regarding base techniques. Overall, these results depict the PSA-TDGN approach as a method that provides improved energy management and system balance relative to hybrid methods evaluated.

Table 5 shows that the proposed TDGNN outperforms all other models under high EV penetration, achieving the lowest MAE and RMSE, and the highest R^2 and accuracy. While Recurrent Neural Network (RNN), Temporal Convolutional Network (TCN), Diffusion Convolutional RNN (DCRNN), and Solid-state transformer perform moderately, TDGNN more effectively captures spatiotemporal patterns, providing significantly improved forecasting accuracy and reliability.

Table 6 shows the ablation results of AFOPID controller tuning using different optimization techniques while employing TDGNN as the forecasting method. The PSA-tuned AFOPID control system demonstrates superior dynamic performance, achieving the lowest overshoot, shortest settling time, fastest rise time, and minimal steady-state error. In contrast, Differential Evolution (DE), Particle Swarm Optimization (PSO), and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) result in inferior performance, with DE and CMA-ES exhibiting particularly high overshoot and steady-state error. These results confirm that

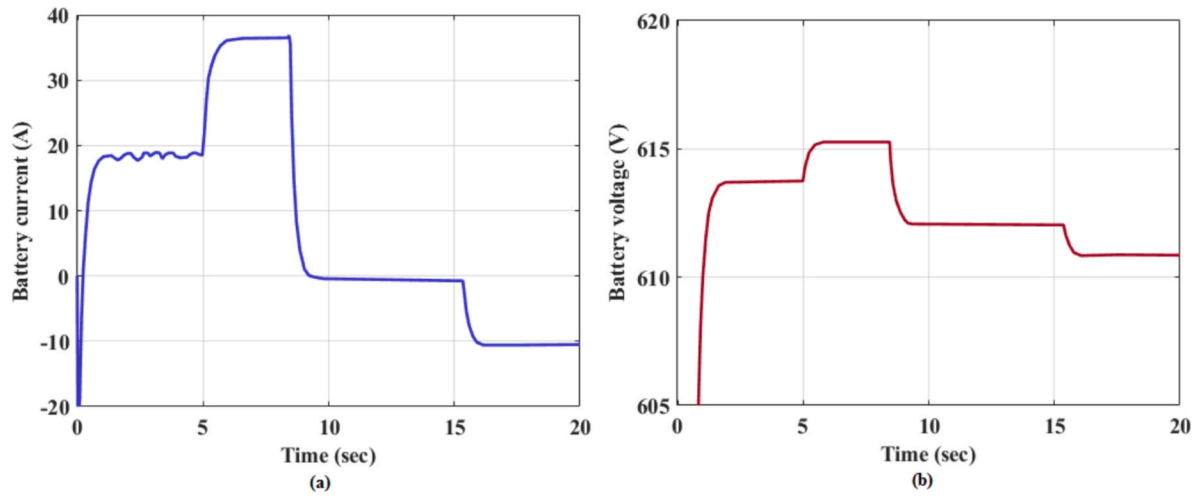


Fig. 8. Analysis of battery operational characteristics (a) Current (b) Voltage.

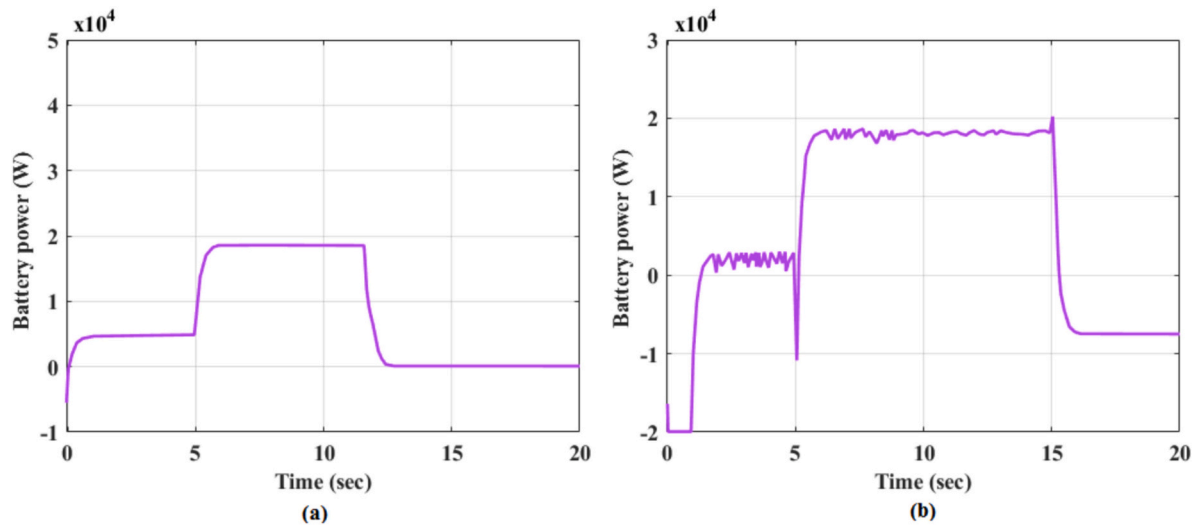


Fig. 9. Performance analysis of energy flow management system with battery during EV charging (a) Grid ON (b) Grid OFF.

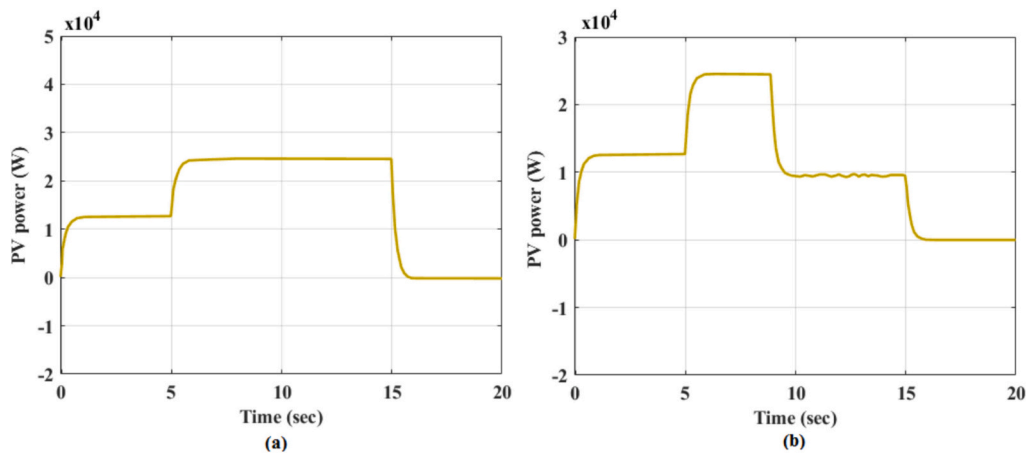


Fig. 10. Performance analysis of energy flow management system with PV during EV charging (a) Grid ON (b) Grid OFF.

PSA, when combined with TDGNN forecasting, provides the most effective and stable controller tuning among the tested approaches.

The ablation results for AFOPID tuning using PSA with various forecasting methods are presented in Table 7. Among them, TDGNN

achieves the lowest energy cost (0.17350\$/kWh), whereas Persistence, LSTM, and GRU result in higher costs. This indicates that AFOPID tuning with PSA is most effective when paired with a robust forecasting method like TDGNN.

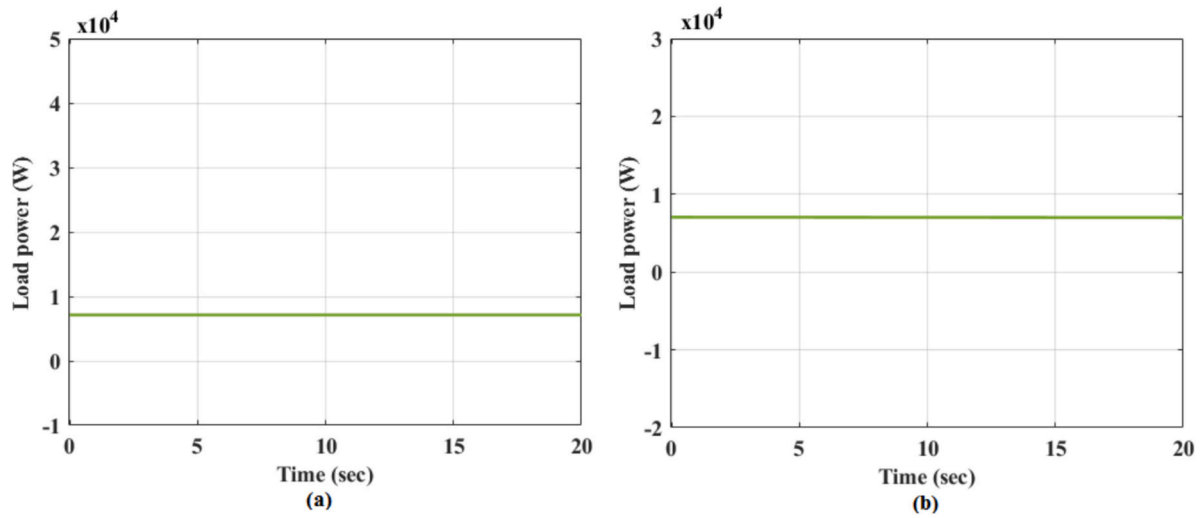


Fig. 11. Performance analysis of energy flow management system with load (a) EV charging, grid ON (b) EV charging, grid OFF.

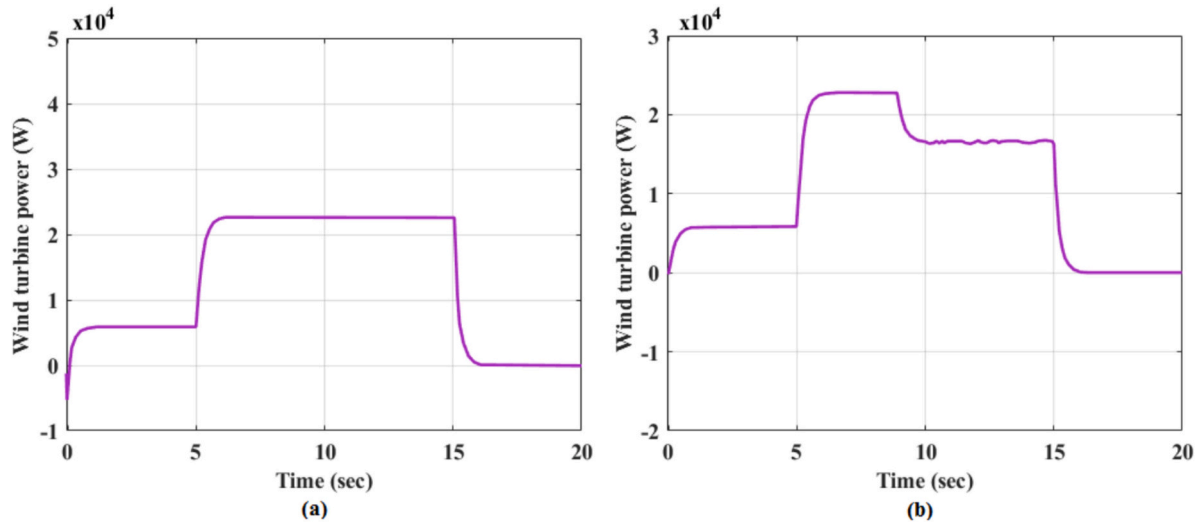


Fig. 12. Performance analysis of energy flow management system with wind power (a) EV charging, grid ON (b) EV charging, grid OFF.

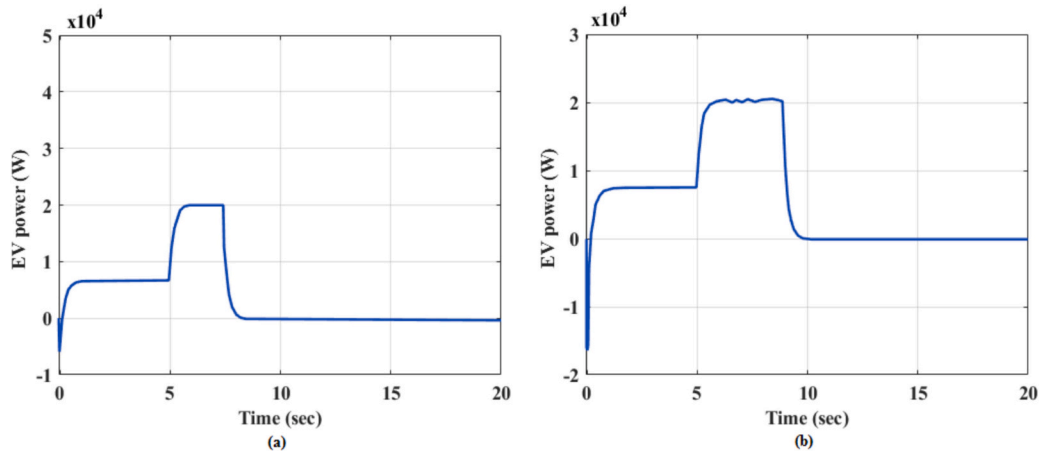


Fig. 13. Performance analysis of energy flow management system with EV charging (a) grid ON (b) grid OFF.

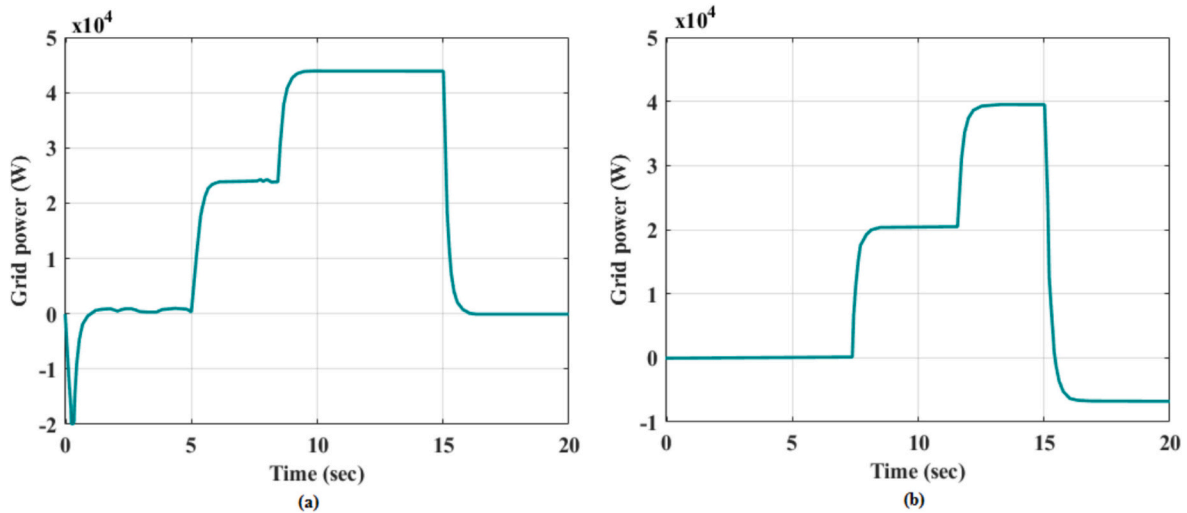


Fig. 14. Performance analysis of energy flow management system grid power (a) grid ON (b) grid OFF.

Table 3

Significance analysis comparing the proposed method with existing approaches.

Methods	Error (%)	Efficiency (%)	Computational complexity (s)	P-value (proposed vs. existing)	Significant ($P < 0.05$)?
a-SCA	2.17	71	56.78	<0.001	Yes
ASSA	1.95	74	42.04	<0.001	Yes
BES	1.26	89	23.04	<0.001	Yes
Proposed	0.59	97	18.23	–	–

Table 4

Comparative analysis of battery SoC and system robustness with previous hybrid techniques.

Reference	Methods	SoC (%)	Robustness (%)
Kranthikumar et al. [18]	EMCABN-ROA	45.09	68
AlKawak et al. [19]	NBO-QNN	55.471	72
Xiaoluan et al. [21]	HCS-PS	81	80
Proposed study	PSA-TDGNN	98.52	87

Table 5

Forecasting performance comparison of TDGNN with other models under high EV penetration.

Model	MAE	RMSE	R ²	Accuracy (%)
RNN	0.0076	0.1316	0.675	79.50
TCN	0.0063	0.1255	0.712	81.95
DCRNN	0.0069	0.1205	0.765	85.5
Solid-state transformer	0.0071	0.9865	0.798	92.67
Proposed	0.0032	0.0823	0.918	97.72

Table 6

Ablation results of AFOPID tuning using different optimization methods with TDGNN forecasting.

Controller tuning	Overshoot	Settling time	Rise time	Steady-state
AFOPID + PSA (proposed)	0.423	0.0132	0.0154	0.01
AFOPID + PSO	2.29	0.0157	0.0187	0.017
AFOPID + DE	11.72	0.0197	0.0246	0.353
AFOPID + CMA-ES	21.04	0.1384	0.0297	0.486

Table 7

Ablation results of AFOPID tuning using PSA with different forecasting methods.

Forecasting method	Cost of energy (\$/kWh)
TDGNN (proposed)	0.17350
Persistence	0.37569
LSTM	0.53678
GRU	0.77450

Table 8

Performance comparison of different EMS methods.

EMS method	Best	Worst	Mean	Standard deviation
MPC-based EMS	9.363	9.465	8.465	2.892
PSO-based EMS	8.696	8.645	7.374	2.546
DE-based EMS	6.126	6.753	6.138	1.845
DQN-based EMS	4.636	5.645	4.826	0.193
Proposed PSA-TDGNN-EMS	3.178	3.432	3.256	0.118

Table 8 compares the performance of several EMS strategies using their best, worst, mean, and standard-deviation values. The results indicate that PSA-TDGNN is the best of all methods, consistently scoring the lowest across all metrics, indicating a more stable and efficient performance. EMS using Deep-Q-Networks (DQN), which is a reinforcement learning (RL) method, is superior to Model Predictive Control (MPC), PSO, and DE, but not to PSA-TDGNN. MPC has the highest values, indicating the least control and more variability overall. Overall, the table establishes that PSA-TDGNN based EMS offers superior and

Table 9

Analysis of per-step runtime for TDGNN and PSA under varying EV counts and prediction horizons.

EV count	Prediction horizon (time steps)	THDGNN prediction time per step (ms)	PSA optimization step time (ms)
10	5	12.5	5.2
10	10	13.2	5.4
50	5	22.8	6.1
50	10	25.4	6.8
100	5	45.1	8.5
100	10	52.3	9.2

Table 10

Analysis of TDGNN performance under out-of-distribution and stress-test scenarios.

Scenario	EV count	SoC range (%)	PV irradiance variation (%)	Wind speed variation (%)	MSE
Normal operation	50	20–80	±10	±10	0.0068
High EV surge	100	20–80	±10	±10	0.0090
Low SoC	50	10–20	±10	±10	0.0079
PV fluctuation	50	20–80	±30	±10	0.0086
Wind fluctuation	50	20–80	±10	±30	0.0087
Combined stress	100	10–20	±30	±30	0.0103

more credible EMS performance.

Table 9 shows that TDGNN prediction time and PSA optimization step time increase with higher EV counts and longer prediction horizons. TDGNN dominates runtime, ranging from 12 to 52 ms, while PSA remains faster, ranging from 5 to 9 ms, indicating reasonable scalability for real-time operation.

Table 10 shows TDGNN performance under normal and stress-test scenarios. The MSE increases with more extreme conditions, such as high EV surge or combined PV and wind fluctuations, but the model remains robust and reliable across all tested operating ranges.

4.1. Discussion

The proposed PSA-TDGNN framework significantly increases the performance of energy management for multi-source microgrids with heavy EV penetration. By combining a PSA-optimized AFOPID controller with a TDGNN-based forecasting model, the proposed framework is capable of controlling energy flows with accuracy and use renewable sources efficiently, while enabling a smooth charging and discharging process for EVs. Simulation results indicate that the proposed framework outperforms conventional EMS frameworks, with the least control error of 0.59%, greatest efficiency of 97%, and fast computation times, while the battery SoC reaches 98.52% and displays generally good system stability. TDGNNs, with their excellent performance on spatiotemporal predictions, are also able to forecast PV and wind generation, load demand, and EV behavior, to achieve proactive energy dispatch and reliable functionality for both grid-connected and islanding scenarios. Ablation studies support the PSA and TDGNN combination being essential, hence other optimizers or forecasters would yield increased energy expenditure, and subsequently poorer dynamics performance. Runtime analysis indicated that the PSA-TDGNN framework was robust in the computational sense, scalable, and responsive, prediction and optimization times were also consistent with the microgrid control intervals, demonstrating that it is appropriate for real-time energy management under high EV penetrations. In addition, stress-test scenarios and regular service model evaluations further support the robustness and resilience of the framework, confirming it is suitable for dynamic operating conditions. Overall, the PSA-TDGNN appears to have substantial efficiency, stability, and predictive power as an energy management approach for developing EVs and other asset types as flexible assets that reduce dependence on the grid and support microgrid sustainable objectives, and overall, it is an effective, considered evaluation approach as researchers develop the next generation of smart, renewable-integrated energy systems.

5. Conclusion

The energy management of a multi-source microgrid that was integrated with EV, RES, and an adaptive fractional order proportional-integral-derivative controller was examined in this study using the PSA-TDGNN technique. The combination of PSA and TDGNN effectively addresses the challenges like error, system efficiency, and system performance. By utilizing the proposed method, it improves the efficiency of the system, reduces errors, and better the performance of the system. The proposed technique is contrasted and accessed utilizing the

MATLAB platform with other existing methods. The proposed strategy outperforms a-SCA, ASSA, and BES in all aspects. The proposed technique achieves an error of 0.59%, which is considerably lower than the errors of a-SCA as 2.17%, ASSA as 1.95%, and BES as 1.26%, indicating superior accuracy in system prediction. In terms of efficiency, the PSA-TDGNN method reaches 97%, far surpassing the efficiencies of a-SCA as 71%, ASSA as 74%, and BES as 89%, ensuring more reliable and stable energy management. These consequences display the effectiveness of the proposed PSA-TDGNN technique in enlightening system efficiency, minimizing grid dependency, and optimizing the total energy management of microgrids. This work is limited to MATLAB-based simulations with a simplified microgrid, and real-time feasibility regarding scalability, delays, and computational overhead remains untested. Future efforts will concentrate on hardware-in-the-loop validation and advanced hybrid meta-heuristics to improve robustness and adaptability.

CRedit authorship contribution statement

A. Madhan: Writing – original draft. **A. Shunmugalatha:** Supervision. **A.S. Vigneshwar:** Supervision.

Consent to participate

Not related.

Consent for publication

Not related.

Ethical approval

None of the authors' studies involving human subjects are included in this article.

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Declaration of competing interest

Not related.

Acknowledgement

None.

Data availability

Data sharing is not applicable to this article because no new data has been created or analysed for this study.

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